

Juzheng Zhang

✉ juzheng@umd.edu | 🏠 juzhengz.github.io | 📧 juzhengz | 🌐 juzheng-zhang | 🐦 juzheng_z | 🎓 Juzheng Zhang

Education

University of Maryland, College Park

PH.D. STUDENT IN COMPUTER SCIENCE

College Park, MD

Aug 2024 – Present

- Advised by [Prof. Tom Goldstein](#); GPA: 4.0/4.0.
- Research on post-training methods for LLMs and alternative generative modeling paradigms for language.

Shanghai Jiao Tong University

BACHELOR OF ENGINEERING IN INFORMATION ENGINEERING

Shanghai, China

Sep 2019 – Jun 2023

- Graduated with Honors (Top 1%); Overall GPA: 4.0/4.3; Rank: 1/152; Member of the Zhiyuan Honors Program.

Publications

LoRI: Reducing Cross-Task Interference in Multi-Task Low-Rank Adaptation

COLM 2025

Juzheng Zhang, Jiacheng You, Ashwinee Panda, Tom Goldstein

HIGHT: Hierarchical Graph Tokenization for Graph-Language Alignment

ICML 2025

Yongqiang Chen, Quanming Yao, Juzheng Zhang, James Cheng, Yatao Bian

UniMoT: Unified Molecule-Text Language Model with Discrete Token Representation

arXiv

Juzheng Zhang, Yatao Bian, Yongqiang Chen, Quanming Yao

Heuristic Learning with Graph Neural Networks: A Unified Framework for Link Prediction

KDD 2024

Juzheng Zhang, Lanning Wei, Zhen Xu, Quanming Yao

Customized Subgraph Selection and Encoding for Drug-drug Interaction Prediction

NeurIPS 2024

Haotong Du, Quanming Yao, Juzheng Zhang, Yang Liu, Zhen Wang

Experience

University of Maryland, College Park

College Park, MD

GRADUATE ASSISTANT

Aug 2024 – Present

- Developed **LoRI**, a LoRA variant using frozen projections and sparse masks to reduce cross-task interference in model merging and continual learning, achieving better performance than LoRA with 95% fewer parameters in single-task and multi-task settings. (COLM 2025)
- Proposed **LoRAF** to mitigate catastrophic forgetting in continual learning using frozen projections and non-overlapping sparse masks, significantly improving knowledge retention in continual learning. (ICLR 2025 SLLM Workshop)
- **Ongoing Research**
- Investigating the limitations of Reinforcement Learning with Verifiable Rewards (RLVR) that cause it to shrink the LLM's reasoning boundary. This involves developing novel strategies to improve sample efficiency and the pass@k metric.
- Developing a reinforcement learning framework for diffusion language models that formulates token unmasking as a sequential decision-making problem, where policies use token-level signals to adaptively control unmasking during the sampling process.

Tsinghua University

Beijing, China

RESEARCH ASSISTANT

Oct 2023 – May 2024

- Developed **UniMoT**, a unified molecule-text language model using a VQ-driven tokenizer with Q-Former for discrete molecule tokens, enabling multi-modal molecule-text generation and comprehension tasks. (arXiv)
- Co-developed **HIGHT**, a hierarchical graph tokenization strategy (node, motif, graph levels) improving graph-language alignment and reducing hallucination by 40% in molecule-language tasks. (ICML 2025)
- Contributed to **CSSE-DDI**, applying Neural Architecture Search principles to automatically discover optimal subgraph selection and encoding functions for improved drug-drug interaction prediction. (NeurIPS 2024)

Baidu Research

Beijing, China

RESEARCH INTERN

Mar 2023 – Oct 2023

- Developed **HL-GNN**, a unified and efficient GNN framework for link prediction that generalizes graph heuristics via matrix formulation; achieved state-of-the-art results on link prediction benchmarks. ([KDD 2024](#))

Honors & Awards

2024	Dean's Fellowship	College Park, MD
2023	Outstanding Graduate of Shanghai Jiao Tong University	Shanghai, China
2021	Shanghai Government Scholarship	Shanghai, China
2021	Merit Student Award	Shanghai, China
2021	Academic Excellence Scholarship	Shanghai, China
2021	Meritorious Winner, International Mathematical Contest in Modeling (MCM)	USA

Services

Reviewer	NeurIPS 25, ICML 25 & 24, ICLR 26 & 25 & 24, KDD 24, and AAAI 26
Teaching Assistant	CMSC 250 Discrete Structures, Fall 2024

Skills

Programming	Python, Pytorch, C/C++, Tensorflow, Verilog
Tools	Git, Docker, Markdown, LaTeX, HTML, CSS
Languages	English, Mandarin